IMEN266 Operations Research II
Chapter 4. Markov Chains

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INDUSTRIAL AND MANAGEMENT
ENGINEERING, POSTECH

Outline

How to study this chapter (2026 format)

- **Slides** (this file): concepts + class-problem statements. No transcription needed — everything typed lives in the Companion.
- **Notebooks** (CP3 modeling · CP4 transient · CP5 limiting/cost · PageRank): in class the `INCLASS` version (fill the `TODOs`, `check()` answers instantly); at home the full version with solutions, sliders, experiments.
- **Companion** (gap-fill → full): all definitions and theorems + 8 core proofs with blanks, checkpoints, AI study cards.
- **Homework 2**: A exam-style (no AI first) · B critique an AI solution · C stress-test the Markov assumption · D prompt log.
- Class rhythm (75 min): quiz $5'$ → concepts → live notebook → wrap-up. Ch. 4 = weeks 3–5.
- Ground truth = analytic ↔ simulation agreement — not AI confidence. Materials: PLMS and GitHub `imen266-2026`.

What is it?

- Stochastic + Process: $\{X_t, t \in \mathcal{I}\}$.
- X_t: random variable for a fixed t.
- $\mathcal{I}$: index set

Random variable \ Index set	Discrete	Continuous
Discrete		
Continuous		

▷ Companion §1.1: the taxonomy table — filled in during the CP3 warm-up.

Class problem 3

1. A company manufactures a machine that is either up or down. If it is up at the beginning of a day, then it is up at the beginning of the next day with probability 0.98 (regardless of the history of the machine); or it fails with probability 0.02. Once the machine goes down, the company sends a repair person to repair it. If the machine is down at the beginning of a day, it is down at the beginning of the next day with probability 0.03 (regardless of the history of the machine); or the repair completes and the machine is up with probability 0.97. A repaired machine is as good as new. Let X_n be the state of the machine at the beginning of the n^{th} day. Model $\{X_n, n \geq 0\}$ as a DTMC.

▶ Colab notebook: CP3 §1 (machine up/down)

Markov chains (MCs, DTMCs)

- $\{X_n, n \in \mathbb{Z}+\}$ is called a Markov chain if it satisfies the following condition:

$$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0)$$

$$= P(X_{n+1} = j | X_n = i)$$

$$= P(X_1 = j | X_0 = i).$$

▷ Companion §1.2 and Proof 1: Markov property ⇒ path probabilities multiply.

Modeling a system as a MC

1. Define X_n, the state at n^{th} observation.
2. Write down $\mathcal{S}$, the state space.
3. Verify Markov & time-homogeneity properties
4. Derive P, the transition probability matrix.
5. Draw a transition diagram.

▷ Companion §1.2 (recipe). Every CP3 problem follows these five steps; `DTMC(P, states)` checks the row sums.

Class problem 3

2. The weather in the city of Pohang is classified as sunny, cloudy or rainy. Suppose that tomorrow's weather depends only on today's weather as follows: If it is sunny today, it is cloudy tomorrow with probability 0.3 and rainy with probability 0.2; if it is cloudy today, it is sunny tomorrow with probability 0.5 and rainy with probability 0.3; and finally, if it is rainy today, it is sunny tomorrow with probability 0.4 and cloudy with probability 0.5. Let X_n be the type of the n^{th} day. Model $\{X_n, n \geq 0\}$ as a DTMC.

▶ Colab notebook: CP3 §2 (Pohang weather)

Class problem 3

3. Himart stocks a wide variety of PCs for retail sales. It is open for business Monday through Friday 8am to 5pm. It uses the following operating policy to control the inventory of Paint-yum PCs: At 5pm Friday, the store clerk checks to see how many Paint-yum PCs are still in stock. If the number is less than 2, then he orders enough Paint-yum PCs to bring the stock up to 5 at the beginning of the business day Monday. If the number in stock is 2 or more, no action is taken. The demand for Paint-yum PCs during the week is a Poisson random variable with mean 3. Any demand that cannot be immediately satisfied is lost. Develop a DTMC to model the inventory of Paint-yum PCs at Himart. For your convenience, use the table below for weekly demand distribution:

k	0	1	2	3	4
$P[\text{Demand} = k]$	0.0498	0.1494	0.2240	0.2240	0.1680
$P[\text{Demand} \geq k]$	1.0000	0.9502	0.8008	0.5768	0.3528

▶ Colab notebook: CP3 §3 (Himart inventory)

Class problem 3

4. There are two identical routes for messages to traverse from node A to node B. If a route was congested when a message needed to be sent, then it will be congested with probability q when the next message needs to be sent. Likewise, if a route was congestion-free when a message needed to be sent, then it will be congestion-free with probability p when the next message needs to be sent. The two routes behave independently. Let X_n be the number of routes available when the n^{th} message is attempted. Show that $\{X_n, n \geq 0\}$ is a DTMC.

▶ Colab notebook: CP3 §4 (two routes)

Class problem 3

5. An individual possesses r umbrellas, which she employs in going from her home to office and vice versa. If she is at home (the office) at the beginning (end) of a day and its raining, then she will take an umbrella with her to the office (home), provided there is one to be taken. If it is not raining, then she never takes an umbrella. Assume that, independent of the past, is rains at the beginning (end) of a day with probability p. Define a DTMC with $r+1$ states that will help us determine the proportion of time that our individual gets wet. Note: She gets wet if it is raining and all the umbrellas are at her other location.

▶ Colab notebook: CP3 §5 (umbrellas)

Class problem 3

6. Consider a machine tool that is used on an NC-machine. The time to produce a part and inspect it is 1-hour. If the tool has produced i parts, it fails with probability p_i. Let X_n be the age of a machine tool at the beginning of n^{th} hour. Assuming that a failed tool is replaced by a new tool whenever it fails, model $\{X_n, n \geq 0\}$ as a DTMC.

▶ Colab notebook: CP3 §6 (tool age)

Class problem 3

7. Consider the following weather forecasting model: if today is sunny (rainy) and it is the k^{th} day of the current sunny (rainy) spell, then it will be sunny (rainy) tomorrow with probability $p_k(q_k)$ regardless of what happened before the current sunny (rainy) spell started. Model this as a DTMC.

▶ Colab notebook: CP3 §7 (weather spells)

Class problem 4

1. Consider the 3-state DTMC for weather in the city of Pohang in the previous class problem. Answer the following questions (all independent):
   (a) According to weather forecast today will be rainy with probability 0.3 and sunny with probability 0.4. What is the probability it would be cloudy tomorrow? How about day after? What about 3 days from today?
   (b) You know that today is sunny. What is the probability that it would be rainy day after tomorrow?
   (c) If yesterday was sunny, day before was sunny and today is rainy, what is the probability that it would be cloudy day after tomorrow?

▶ Colab notebook: CP4 §1 (weather, transient)

Transient analysis

- Initial distribution

$$a \equiv [P(X_0 = 0), P(X_0 = 1), \ldots] .$$

- Define

$$p_n \equiv [P(X_n = 0), P(X_n = 1), \ldots] .$$

- Transient analysis

$$\begin{aligned} p_1 &= aP \\ p_2 &= p_1 P = aP^2 \\ &\vdots \\ p_n &= aP^n . \end{aligned}$$

▷ Derivation: Companion, Proof 2 (Chapman–Kolmogorov, $p_n = aP^n$).

Class problem 4

3. Consider a DTMC with initial distribution a given by $a = (0.3, 0, 5, 0.1, 0.1)$ and transition probability matrix P given by

$$P = \begin{pmatrix} 0.25 & 0.25 & 0.25 & 0.25 \\ 0.1 & 0.2 & 0.3 & 0.4 \\ 0.6 & 0.1 & 0.1 & 0.2 \\ 0.2 & 0.2 & 0.2 & 0.4 \end{pmatrix}.$$

Compute the following:

(a) $P[X_3 = 3, X_2 = 3, X_1 = 2, X_0 = 1]$
(b) $P[X_3 = 3, X_2 = 2, X_1 = 1]$
(c) $P[X_5 = 2 | X_1 = 3]$
(d) $P[X_3 = 1]$
(e) Distribution of X_5
(f) $E[X_4]$
(g) Var$[X_4]$

▶ Colab notebook: CP4 §3 (4-state DTMC)

Class problem 4

2. Consider the 4-state DTMC for the Himart inventory PCs problem in the previous class problems. Assume that the business starts with 5 Paint-yum PCs in hand at the beginning. Answer the following questions (all independent):
   (a) What is the initial distribution a?
   (b) What is the probability that at the end of that week Paint-yum PCs have to be ordered for the next week?
   (c) What is the probability that after 4 weeks there are 3 Paint-yum PCs?
   (d) If you are given that at the beginning of the 17th week there are 3 Paint-yum PCs, what is the probability that at the beginning of the 20th week there would be 5?

▶ Colab notebook: CP4 §2 (Himart, transient)

Important properties

$$P = \begin{pmatrix} 0.6 & 0.4 & 0 & 0 & 0 \\ 0.8 & 0.2 & 0 & 0 & 0 \\ 0 & 0 & 0.3 & 0.2 & 0.5 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0.8 & 0.1 & 0.1 \end{pmatrix}$$

$$P = \begin{pmatrix} 0.6 & 0.4 & 0 & 0 & 0 \\ 0.3 & 0.7 & 0 & 0 & 0 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 \\ 0 & 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0 & 0.5 & 0.5 \end{pmatrix}$$

Important properties

- Irreducible vs. reducible
- Recurrent vs. transient
- Aperiodic vs. periodic

▷ Definitions: Companion §1.4; Proof 3 (recurrence criterion, class property); `DTMC.summary()` classifies for you.

Desired Markov chains (Ergodic)

- Irreducible
- Positive recurrent
- Aperiodic

▷ Companion §1.4: ergodic = irreducible + positive recurrent + aperiodic. Proof 7: why taxation makes PageRank ergodic.

Limiting behavior

- If a Markov chain is irreducible and positive recurrent, we have a *unique solution* to

$$\boldsymbol{\pi} = \boldsymbol{\pi} P, \quad \sum_{i \in \mathcal{S}} \pi_i = 1,$$

where $\boldsymbol{\pi} = (\pi_0, \pi_1, \ldots)$.

- If the Markov chain above is aperiodic,

$$\lim_{n \to \infty} P(X_n = i) = \pi_i.$$

- Example

▷ Companion §1.5; Proof 4 (two-state $P^n \to \pi$), Proof 5 (solving $\pi = \pi P$), Proof 6 ($\pi_i = 1/m_i$).

Class problem 5

1. Consider problem #1 in Class Problems 3 on DTMC modeling. If it is up at the beginning of a day, then it is up at the beginning of the next day with probability 0.98 (regardless of the history of the machine); or it fails with probability 0.02. Once the machine goes down, the company sends a repair person to repair it. If the machine is down at the beginning of a day, it is down at the beginning of the next day with probability 0.03 (regardless of the history of the machine); or the repair completes and the machine is up with probability 0.97. For this system, compute the steady state probabilities that the machine is up and down at the beginning of a day in the long run.

▶ Colab notebook: CP5 §1 (machine, steady state)

Class problem 5

2. Consider problem #2 in Class Problems 3 on DTMC modeling about the weather being one of sunny, cloudy or rainy everyday. If it is sunny today, it is cloudy tomorrow with probability 0.3 and rainy with probability 0.2; if it is cloudy today, it is sunny tomorrow with probability 0.5 and rainy with probability 0.3; and finally, if it is rainy today, it is sunny tomorrow with probability 0.4 and cloudy with probability 0.5. What fraction of days would be sunny, cloudy or rainy in this city?

▶ Colab notebook: CP5 §2 (weather, steady state)

Class problem 5

3. Consider problem #5 in Class Problems 3 on DTMC modeling about this lady with r umbrellas. Compute the limiting probabilities $(\pi_0, \pi_1, \ldots, \pi_r)$ and also determine the fraction of trips the individual becomes wet (φ). Every time the individual becomes wet, it costs c won to recover (medication, new clothes, or whatever). Also the maintenance cost for r umbrellas is d won per umbrella per trip (basically to purchase new umbrellas to replace old ones). Let τ be the long-run average cost per trip, then $\tau = c\varphi + dr$. In terms of p, c and d, determine the optimal number of umbrellas the individual should possess.

▶ Colab notebook: CP5 §3 (umbrellas, cost)

Cost and reward

- If a cost $C(i)$ is incurred everytime the Makov chain is observed in state j, then the long-run average cost per observation is
$$\sum_{i \in \mathcal{S}} C(i)\pi_i.$$

▷ Proof 6 in the Companion: time average $\to \sum_i C(i)\pi_i$; mean return time $1/\pi_i$.

Class problem 5

4. Consider problem #6 in Class Problems 3 on DTMC modeling where if a machine tool produced i parts, it fails with probability p_i. Let X_n be the age of a machine tool at the beginning of n^{th} hour. Compute the steady state distribution for the DTMC. What is the average age of a tool when it starts processing a part? What is the average lifetime of a tool?

▶ Colab notebook: CP5 §4 (tool age, steady state)

Class problem 5

5. Kim is a student that goes out to eat lunch everyday. He eats either at a Korean food place (0), Chinese food place (1), or Burger place (2). The place Kim chooses to go for lunch on day n can be modeled as a DTMC with transition matrix

$$\begin{pmatrix} 0.3 & 0.3 & 0.4 \\ 1 & 0 & 0 \\ 0.5 & 0.5 & 0 \end{pmatrix}.$$

That means if Kim went to the Korean food place yesterday, he will choose to go to the Burger place today with probability 0.4, also there is a 30% chance he will go to the Chinese food place. Are there any absorption states? Is the DTMC ergodic?

▶ Colab notebook: CP5 §5 (Kim's lunch: model)

Class problem 5

(a) After several days (assume steady state is reached) if you ask Kim what he had for lunch, what is the probability that he would say "burger"?

(b) If on day $n = 1000$ Kim went to the Korean food place, what is the probability that on day $n = 1001$ he will go to the Chinese food place?

(c) If all the students at Kim's university follow the above Markov chain, in the long run, what fraction of the students going to lunch eat at the Chinese food place?

(d) Continuing with question (c) above, if the average cost per lunch is $4,000$ won in the Burger place, $6,000$ won in the Chinese place and $5,000$ won in the Korean place, what is the average revenue per day per customer from all the three eating places together?

(e) How much does Kim spend on lunch (on average) everyday?

▶ Colab notebook: CP5 §5 (Kim's lunch: questions)

PageRank: the random surfer is a Markov chain

- Surfer on n pages: from page i with k links, follow one uniformly $\Rightarrow$ a DTMC with matrix P; the rank of page j is π_j.
- What breaks ergodicity: dead ends (no links out: mass leaks) and spider traps (closed pages: π concentrates there).
- Taxation with rate $1-\beta$: $G = \beta P + \frac{1-\beta}{n}\mathbf{1}\mathbf{1}^\top$ is strictly positive $\Rightarrow$ ergodic, unique π, power iteration converges at rate $\leq \beta$.
- Handout: $\pi = (0.4, 0.4, 0.2)$; with the spider trap and $\beta = 0.7$, $\pi \approx (0.26, 0.19, 0.55)$.

▷ Proof 7 in the Companion (exam topic): ergodicity of G and $\|p_{k+1} - \pi\|_1 \leq \beta \|p_k - \pi\|_1$.

▶ Colab notebook: PageRank practicum (link farm, taxation, exam drill)